

OSAROGIE EXCELLENT OGIEMUDIA

Field Deployment Engineer · Cyber-Physical Systems Specialist · MSc Cyber Security Candidate

 Plymouth, Devon, UK ·  07502 479852 ·  Osarogieernest@gmail.com ·  osarogie-portfolio.vercel.app ·  github.com/Exceluxred · Full UK Right to Work

PROFESSIONAL PROFILE

Multi-disciplinary Field Engineer and Cybersecurity MSc candidate with proven commercial experience deploying, integrating, and securing physical-to-digital infrastructure. Proven expertise across Southwest UK commercial field engineering (racking, stacking, configuring Fortigate firewalls, Cisco switches, structured cabling, and mission-critical Dell server fiber NIC replacements at Derriford Hospital) and autonomous marine robotics platforms (electro-mechanical assembly, component-level PCB soldering, waterproofing, and offshore sea trials in Plymouth Sound at Oshen Ltd). Combines practical mechanical and electrical diagnostic skills with strong computer science fundamentals, NIST SP 800-207 Zero-Trust architecture, and systems programming in C and Python.

CORE TECHNICAL COMPETENCIES

Field & Network Infrastructure:

Fortigate Firewalls, Cisco Managed Switches, Patch Panels, Cat6/Fiber Terminations, PCIe SFP+ NICs, Wireless Access Points (WAPs), ServiceNow SOWs, Rack & Stack.

Hardware, Robotics & Electronics:

Autonomous Surface Vessels (USV), Component-Level Soldering, Wiring Harnesses, Microcontrollers, Hull Waterproofing & Pressure Testing, Server Hardware Diagnostics.

Cybersecurity & Defensive Systems:

Zero-Trust Architecture (NIST SP 800-207), Threat Hunting, Wireshark & PCAP Forensics, Snort & Zeek NIDS, MITRE ATT&CK Mapping, Active Directory, Incident Response.

Systems Programming & Operations:

C (GNU89 / POSIX Systems), Python, Bash Scripting, Linux System Administration (Ubuntu/Debian), Git/GitHub, ITIL® 4 Foundation IT Service Management.

PROFESSIONAL EXPERIENCE

Field Deployment Engineer — Commercial Deployments (Southwest UK) 2024 – Present

- Field Rollouts & Network Delivery:** Conduct on-site network upgrades and hardware migrations across enterprise, public-sector, and retail sites in Devon and Cornwall (Plymouth, Exeter, Truro, Torquay).
- Hardware Integration & Commissioning:** Rack, stack, cable, and patch Fortigate security firewalls, Cisco managed switches, patch panels, and power distribution units (PDUs) adhering to strict vendor SOWs.
- Critical Server Hardware Maintenance:** Successfully executed dual-port PCIe SFP+ fiber optic Network Interface Card (NIC) replacements on live Dell PowerEdge enterprise servers at Derriford Hospital.
- Cabling & Wireless Infrastructure:** Install, terminate, and test Cat6 copper and fiber optic runs; mount, patch, and verify enterprise Wireless Access Points (WAPs) for optimal RF signal coverage.
- Documentation & Ticket Delivery:** Complete structured post-installation testing, capture photo verification deliverables, and secure customer sign-off on ServiceNow ticketing workflows.

Marine Robotics Technician — Oshen Ltd (Plymouth, UK) 2024 – 2025

- Autonomous Robotic Assembly:** Assembled autonomous sailing vessel platforms (C-Star) from structural chassis to operational launch, integrating sensor suites, microcontrollers, wiring harnesses, and battery power systems.
- Component-Level Soldering & Sealing:** Carried out high-precision PCB through-hole and surface soldering, wiring loom crimping, structural hull waterproofing, and airtight pressure testing for open-ocean resilience.
- Sea Trials & Boat Handling:** Participated in live offshore sea trials in Plymouth Sound from support craft; deployed vessels, monitored navigation telemetry and sensor packet streams, and executed vessel recovery.
- Rapid Dockside Troubleshooting:** Completed benchtop diagnostics, rapid electromechanical component swaps, sensor recalibrations, and hull re-sealing between operational deployment windows.

Technical Instructor & Facilitator — New Horizons Computer Learning Centers (Abuja, Nigeria) 2021 – 2024

- Lab Systems Administration:** Managed and maintained hardware operability, OS imaging, structured networking, and troubleshooting for over 50 lab workstations and server environments.
- Hardware & Robotics Mentorship:** Instructed cohorts in computer architecture, hardware diagnostics, sensor integration, microcontroller programming, and applied engineering challenges.
- Infrastructure & Security Support:** Administered on-site network equipment, managed user accounts and access policies in Active Directory, and maintained the hardware repair testbenches.

- **Modular & Board-Level Repairs:** Diagnosed, troubleshoot, and completed hardware repairs across laptops, desktop workstations, power supplies, logic boards, and mobile devices.
- **Systems Maintenance & Imaging:** Deployed customized OS images, resolved peripheral and device driver conflicts, and configured local IP addressing and network shares.
- **Standards Compliance:** Maintained structured repair tracking logs, inventory control, and adhered to CompTIA A+ hardware testing and ESD safety guidelines.

TECHNICAL PROJECTS & SYSTEMS ARCHITECTURE

Zero-Trust Policy Enforcement & Access Controller

github.com/Exceluxred/zerotrust ↗

Architected a NIST SP 800-207 compliant Policy Enforcement Point (PEP) and Policy Decision Point (PDP) engine in Python. Enforces continuous dynamic access verification, token authentication, and micro-segmentation models designed for cyber-physical systems (CPS) and industrial IoT edge platforms.

NIST SP 800-207 • Cyber-Physical Systems • Python • Micro-Segmentation • Network Security

SOC Incident Response & Malware PCAP Analysis

github.com/Exceluxred/soc-incident-response-malware-analysis ↗

Investigated a multi-stage intrusion involving Dridex banking trojan delivery and Remcos RAT command-and-control beaconing. Reconstructed full forensic attack timelines via Wireshark PCAP analysis, crafted Snort and Zeek detection rules, and mapped adversary TTPs to the MITRE ATT&CK framework.

Wireshark • PCAP Packet Forensics • Snort • Zeek • MITRE ATT&CK • Threat Hunting • Python

POSIX UNIX Command Interpreter (C GNU89)

github.com/Exceluxred/simple_shell ↗

Engineered a custom POSIX command interpreter in standard C (GNU89) with process isolation (fork/execve), signal handling, custom tokenizer, and built-in path resolution. Achieved 100% leak-free heap memory verified via Valgrind.

C (GNU89) • UNIX / POSIX API • Process Management • Valgrind Leak-Free • Linux Internals

EDUCATION & CREDENTIALS

MSc Cyber Security

University of Plymouth, United Kingdom
Sept 2024 – Expected 2026

- Dissertation: *Zero Trust Architecture For Securing Cyber-Physical Systems*
- Modules: Network Defense, Digital Forensics, Systems Security, Cryptography

National Diploma (ND) Civil Engineering

Auchi Polytechnic, Nigeria
Sept 2015 – Dec 2018

- Engineering Mechanics, Technical Drafting, Surveying, Structural Quality Assurance

BSc Computer Science

Benson Idahosa University, Nigeria
Sept 2019 – May 2022

- Core: Data Structures & Algorithms, OS Internals, Networking, Computer Architecture
- Foundations: Systems Analysis, Database Management, Object-Oriented Design

ITIL® 4 Foundation in IT Service Management

AXELOS / PeopleCert · Certified
Service Management Lifecycle

- Incident Management, Change Control, Continual Improvement, Service Level Agreements

FRAMEWORKS & OPERATIONAL STANDARDS

Security & IT Service Frameworks: NIST SP 800-207 (Zero Trust) • MITRE ATT&CK Enterprise Matrix • ITIL® 4 IT Service Management • ServiceNow Ticketing & SLA Compliance • CompTIA A+ Hardware Diagnostic Guidelines • POSIX / GNU89 Systems Standards.